# CS & IT ENGINEERING




## COMPUTER ORGANIZATION AND ARCHITECTURE

### Basics of COA

Lecture No.- 02




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# Recap of Previous Lecture



- **Topic** Architecure vs Organization
- **Topic** Numbers & Data in Computers
- **Topic** Components of Computer
  CPU, Mem., I/O
- **Topic** System Buses
- **Topic** Types of Buses

# Topics to be Covered



- Topic: CPU Registers
- Topic: Types of Architecture
- Topic: Program Counter
- Topic: Instruction Register
- Topic: Stack Pointer

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

$2^8 = 256$

$2^9 = 512$

$2^{10} = 1024$

$2^{11} = 2048$

$2^{12} = 4096$

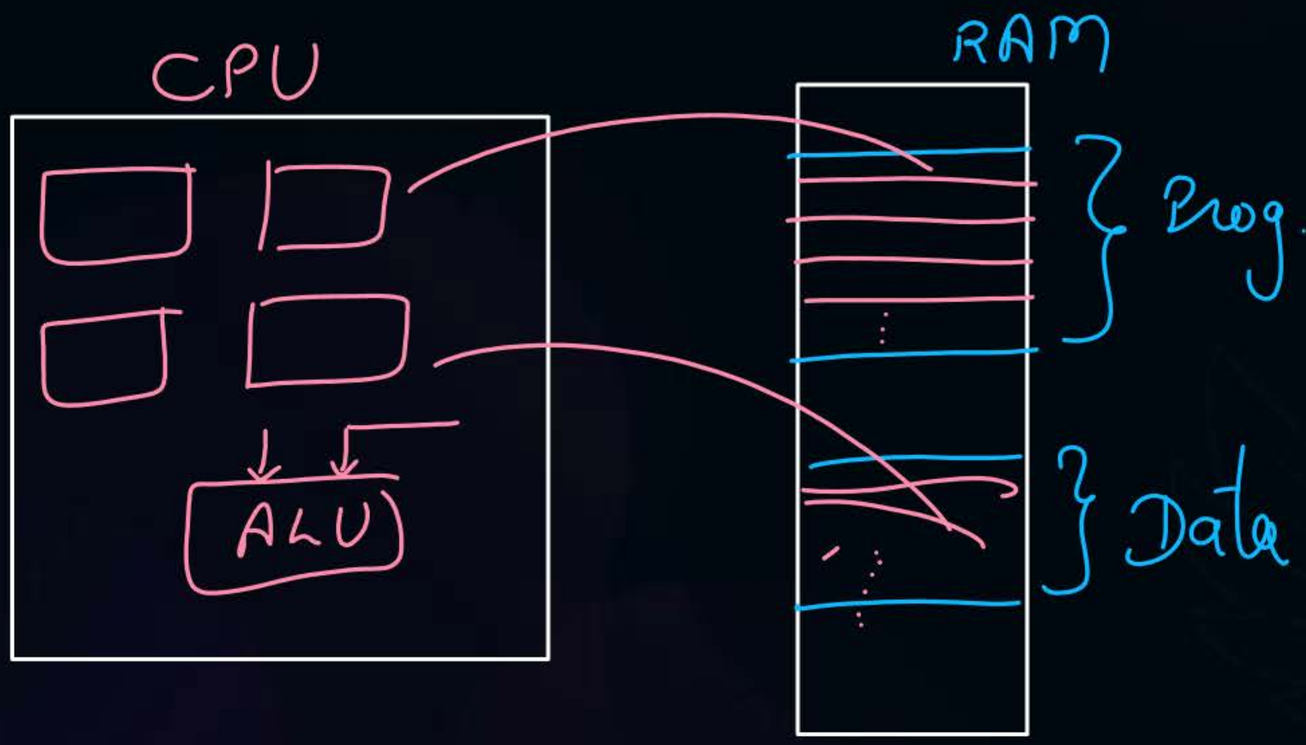
$2^{13} = 8192$

$2^{14} = 16384$



# Topic : CPU Registers

small memories inside CPU to carry out processing.




Reg. Capacity

⇓

8 - bits ⇒ 1 Byte

or

16 bits ⇒ 2 bytes

or

32 bits ⇒ 4 bytes

or

64 bits ⇒ 8 bytes



# Topic : CPU Registers

## CPU Register

- General Purpose Registers (GPRs) ⇒ R0, R1, R2, R3, .....
- Special Purpose Registers



# Topic : CPU Registers

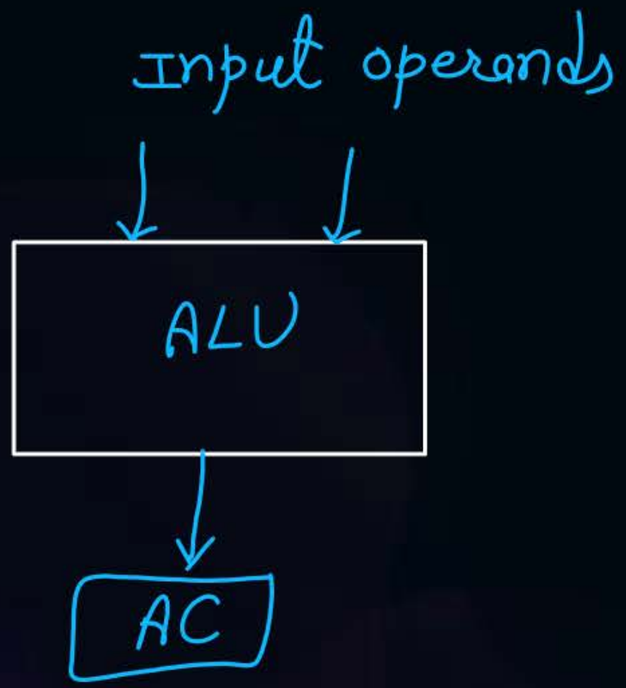
## CPU Register

- General Purpose Registers (GPRs)
- Special Purpose Registers
  1. Accumulator (AC)
  2. Program Counter (PC)
  3. Instruction Register (IR)
  4. Stack Pointer (SP)
  5. Flag Register / Program Status Word (PSW) / status Register
  6. Address Register (AR) / Memory Address Register (MAR)
  7. Data Register (DR) / Memory Data Register (MDR) / MBR
     ↳ Buffer

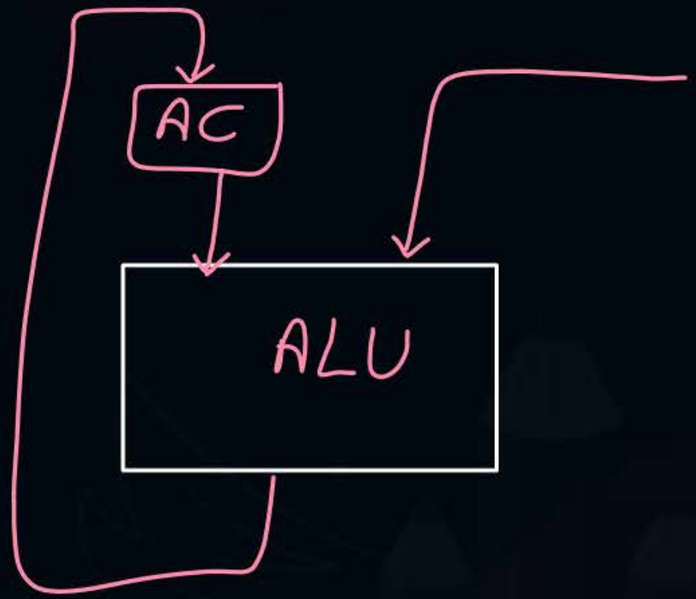


# Topic : Accumulator

- Used to store result of ALU and sometimes one of the operands for ALU too.




$AC \leftarrow a + b$




$AC \leftarrow a$

$AC \leftarrow AC + b$



# Topic : Types of Architecture

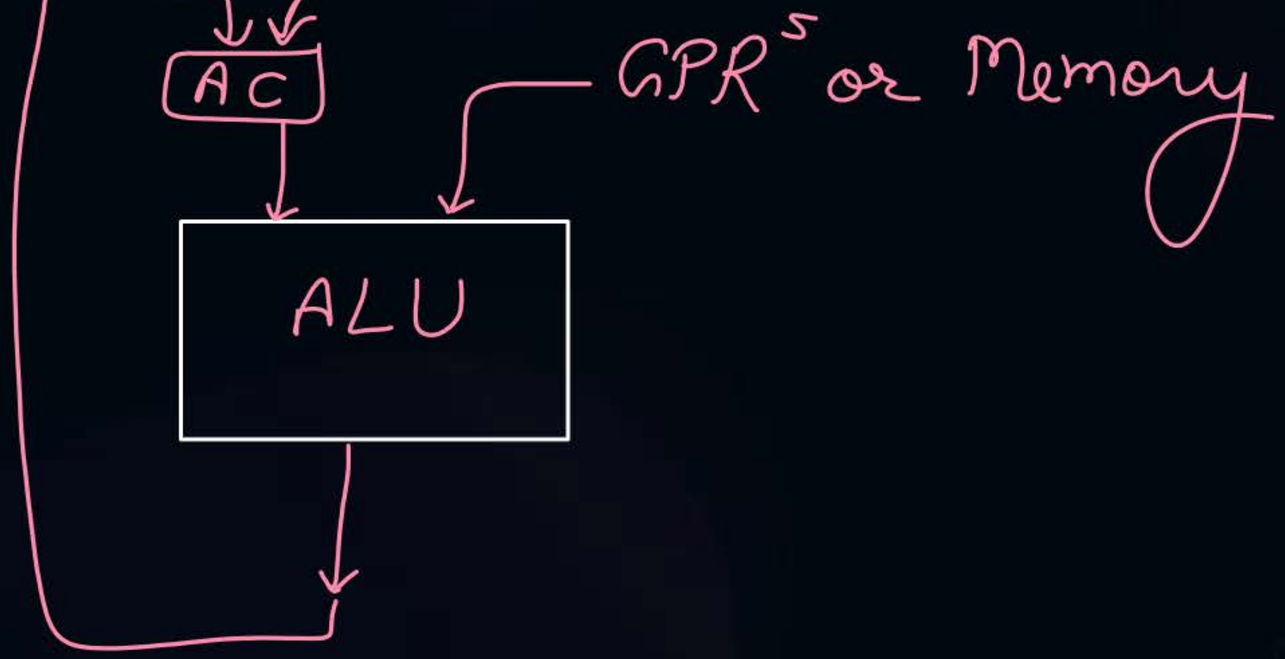
operands

**Based on ALU input:** → from where 2 inputs of ALU can be taken.

- AC-Based Architecture
- Register Based Architecture
- Register-Memory Based Architecture
- Complex System Architecture
- Stack Based Architecture

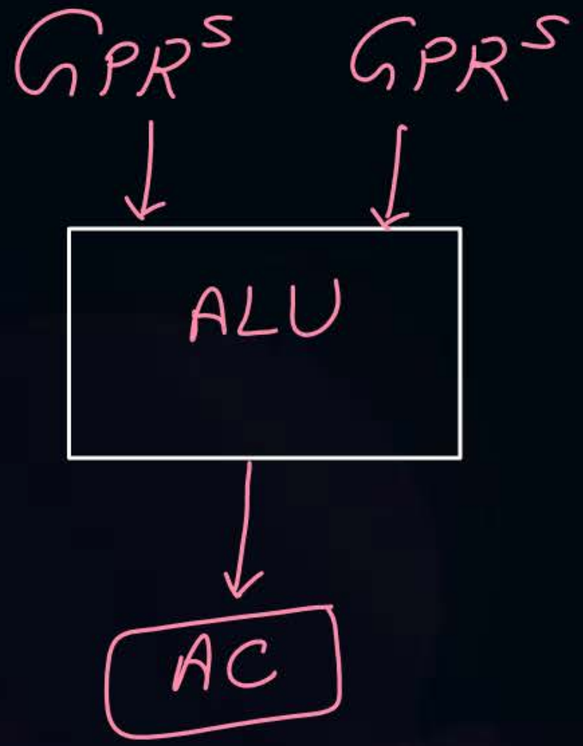


# Topic : AC-Based Architecture





# Topic : Register-Based Architecture




a ⇒ in mem.

b ⇒ in mem.

if we want a + b then first bring a, b in GPRs

⇓

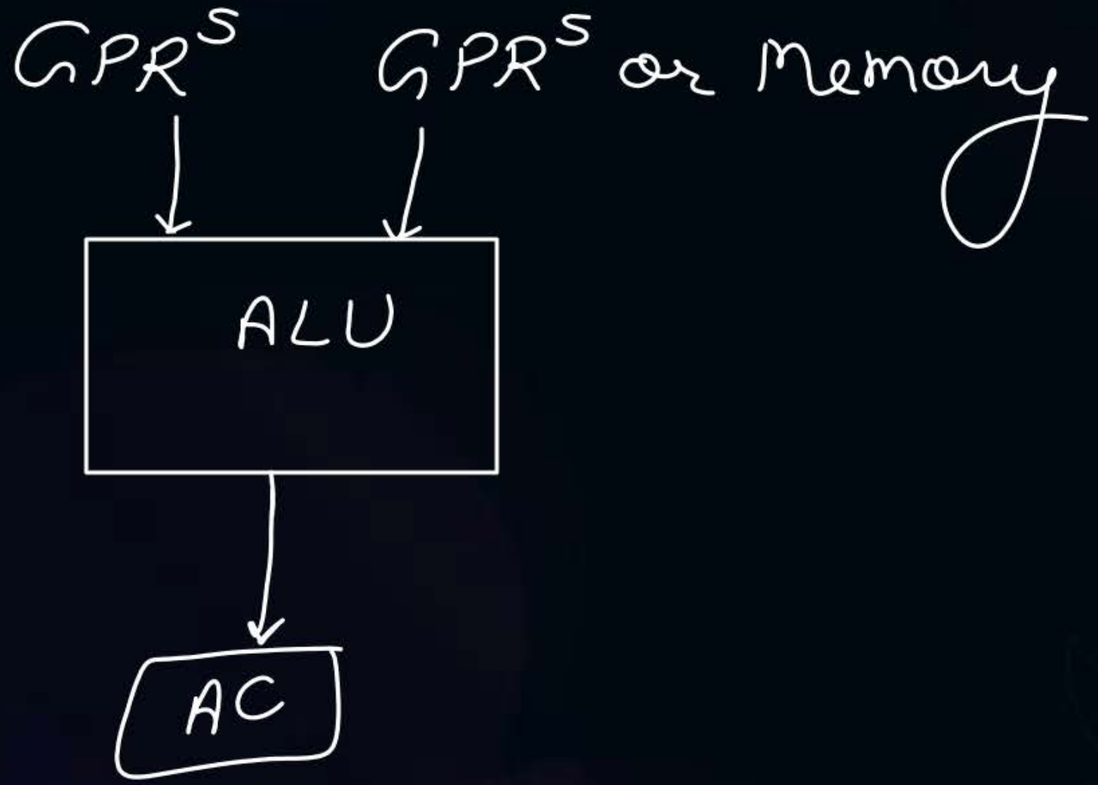
R1 ⟵ a

R2 ⟵ b

AC ⟵ R1 + R2



# Topic : Register-Memory Based Architecture

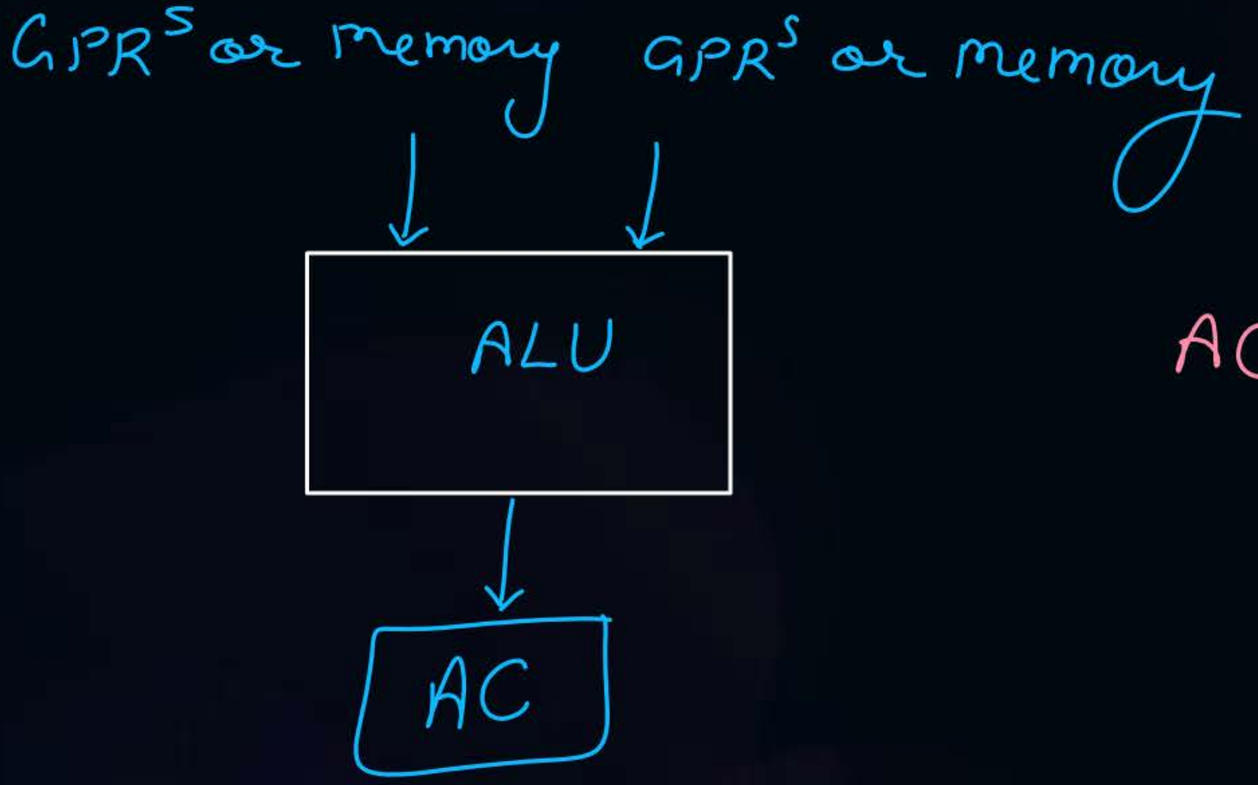



R1 ← a

AC ← R1 + b



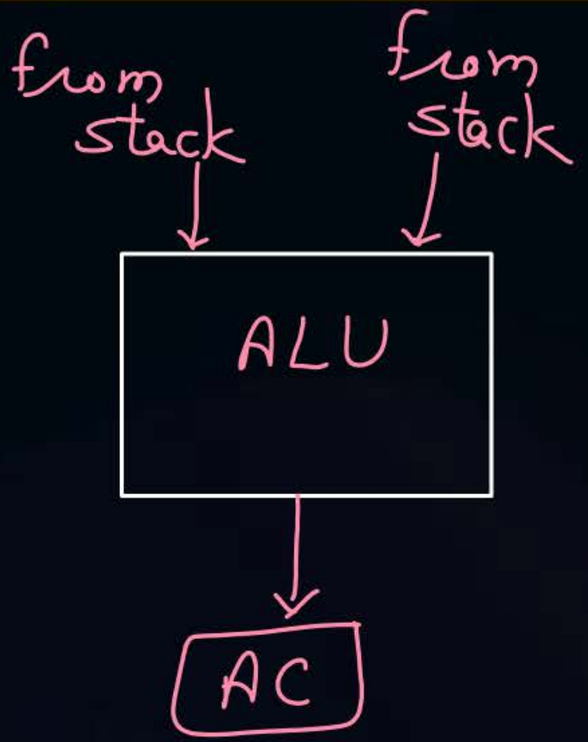
# Topic : Complex System Architecture




$AC \leftarrow a + b$



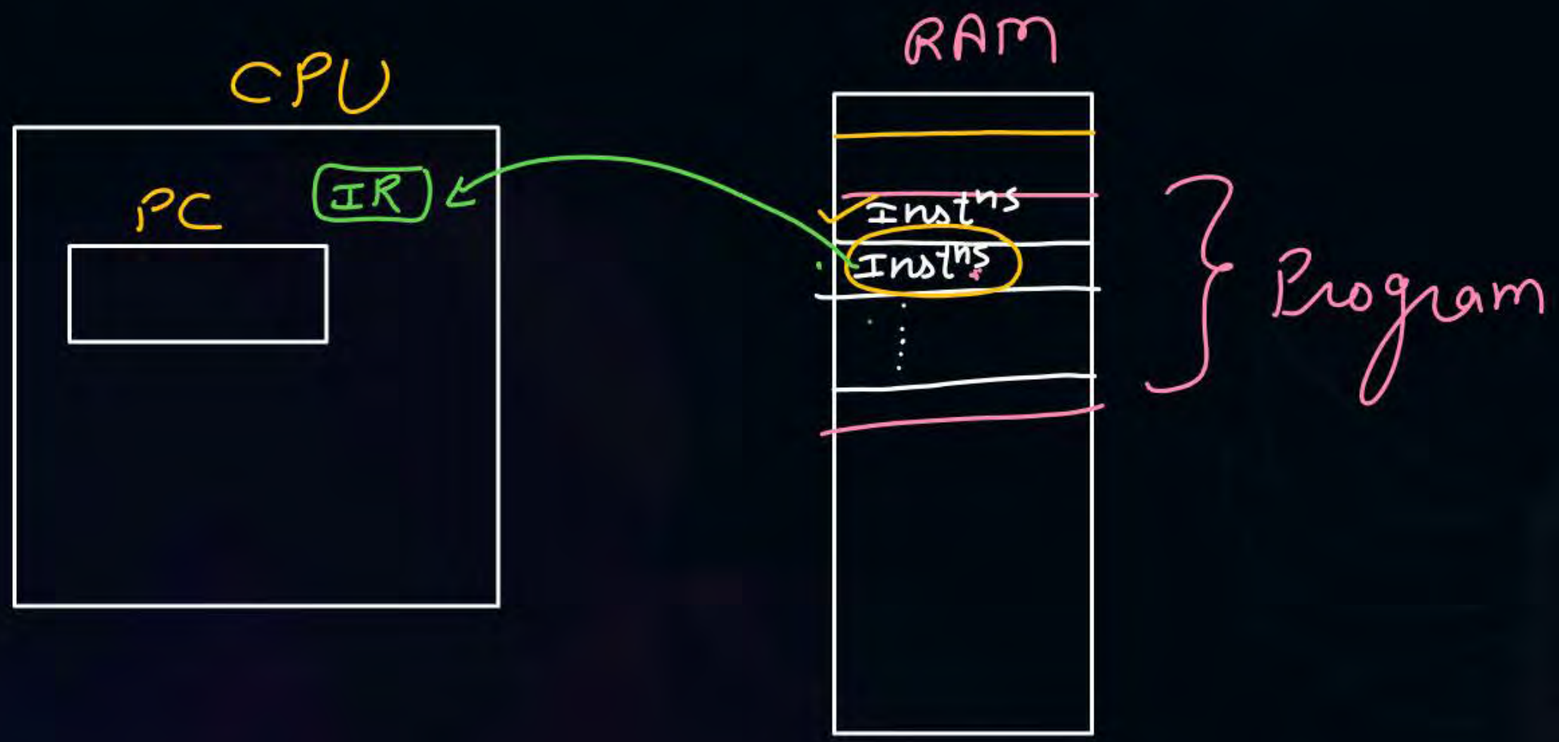
# Topic : Stack-Based Architecture





# Topic : Program Counter

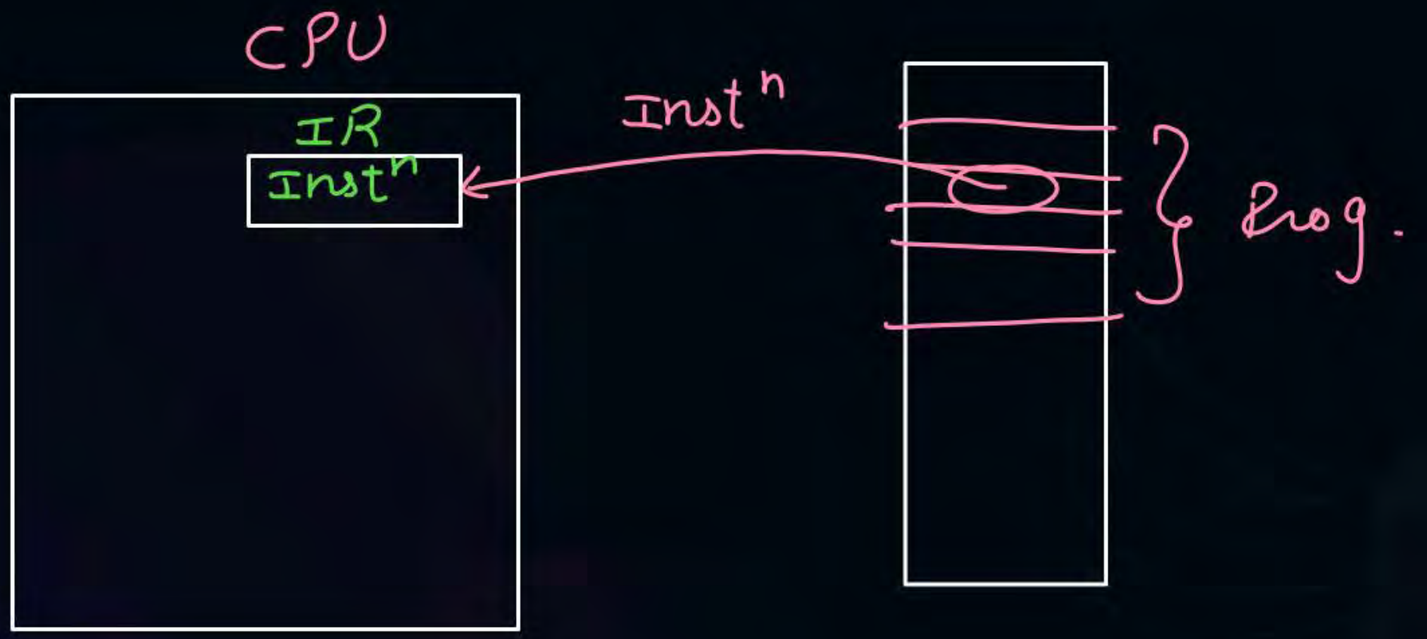
- Stores address of next instruction to be executed





# Topic : Instruction Register

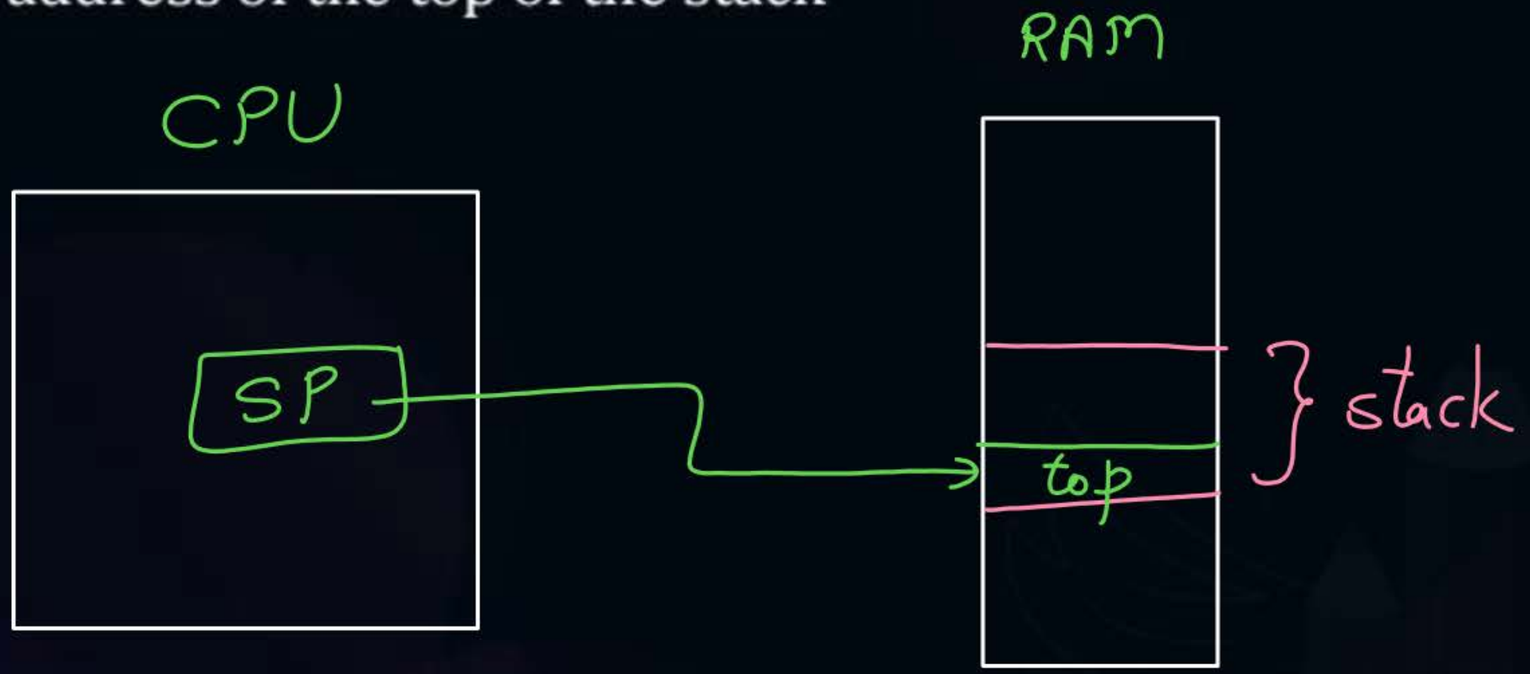
- Stores the current instruction to be executed





# Topic : Stack Pointer

- Stores the address of the top of the stack

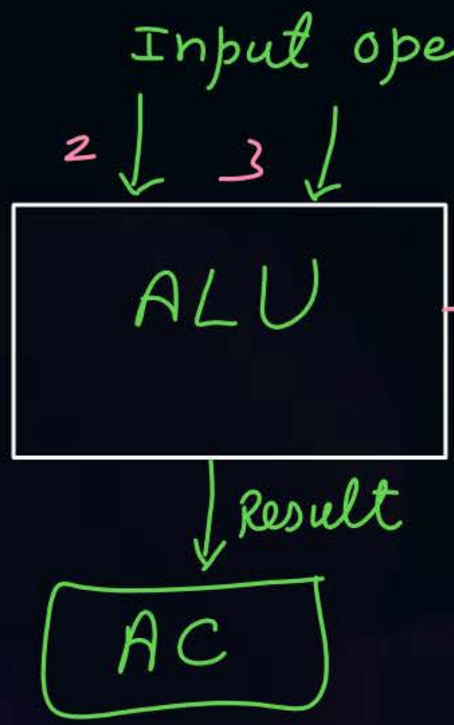




# Topic : Flag or Status Register

- Stores the status of the ALU result

Used for condition checking




status

status Register

| S | Z | Ca | ..... |
|---|---|---|---|

S ⇒ sign { 0 ⇒ +ve, 1 ⇒ -ve

Z ⇒ zero or non-Zero

Ca ⇒ carry

if (a > b)

⇓

CPU ALU ⇒ a − b

check status that result was positive or not.



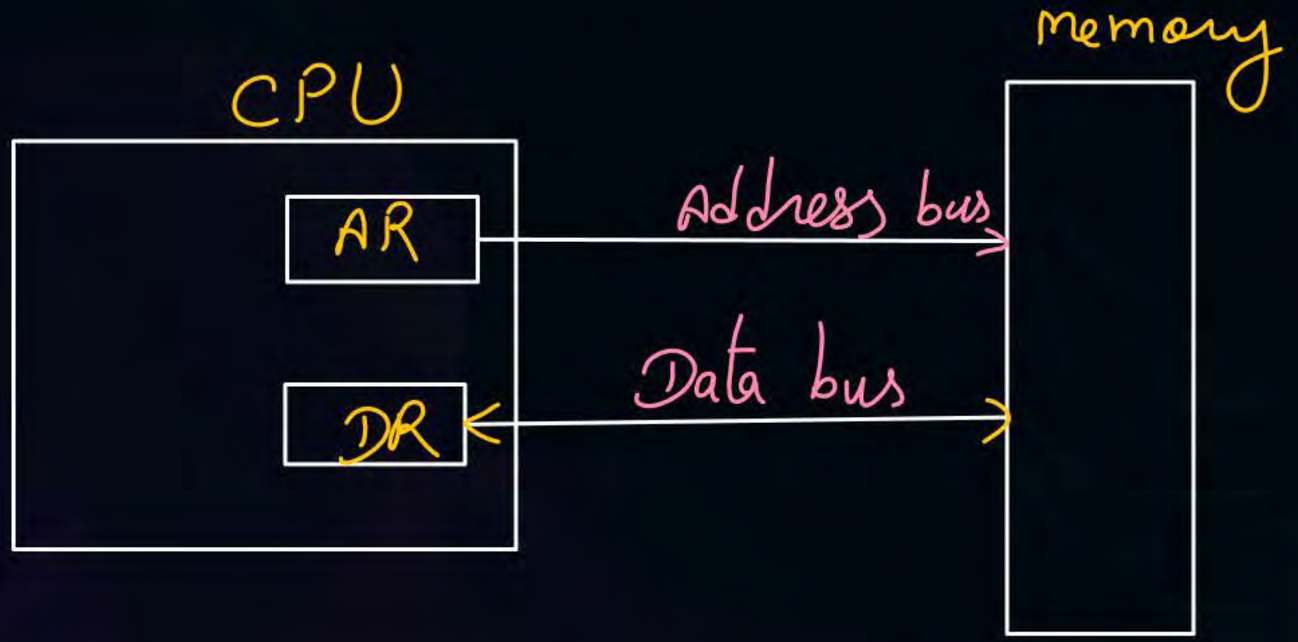
# Topic : Address Register or MAR

- Used to send address to memory



# Topic : Data Register or MDR

- Used to send data to memory
- And to receive data from memory





# 2 mins Summary

- Topic: CPU Registers
- Topic: Types of Architecture
- Topic: Program Counter
- Topic: Instruction Register
- Topic: Stack Pointer



# Happy Learning

# THANK - YOU